SCSI 1013: Discrete Structure

CHAPTER 4

GRAPH THEORY

Definition of Graph

- A graph G consists of two finite sets:
  - a nonempty set $V(G)$ of vertices.
  - a set $E(G)$ of edges, where each edge is associated with a set consisting of either one or two vertices called its endpoints.
  - f is a function, called an incidence function, that assign to each edge, $e \in E$, a one element subset $\{v\}$ or two elements subset $\{v\ ,\ w\}$, where v and w are vertices.
- We can write G as (V, E, f) or (V, E) or simply as G.

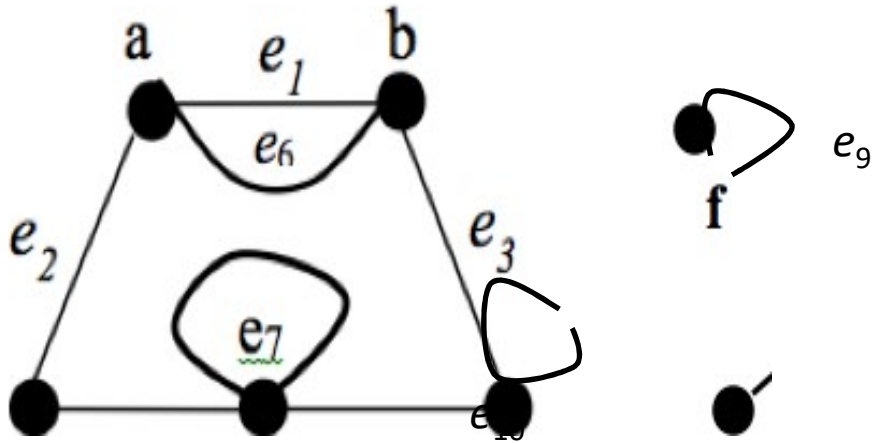
Example 1

- Let,
  - $V = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$
  - $E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$
- And f be defined by:
  - $f(e_1) = f(e_2) = \{v_1, v_2\}$
  - $f(e_3) = \{v_4, v_3\}$
  - $f(e_4) = f(e_6) = f(e_7) = \{v_6, v_3\}$
  - $f(e_5) = \{v_2, v_4\}$

Question: What is the pictorial representation of *G*?

Example 2

Given a graph as shown below,

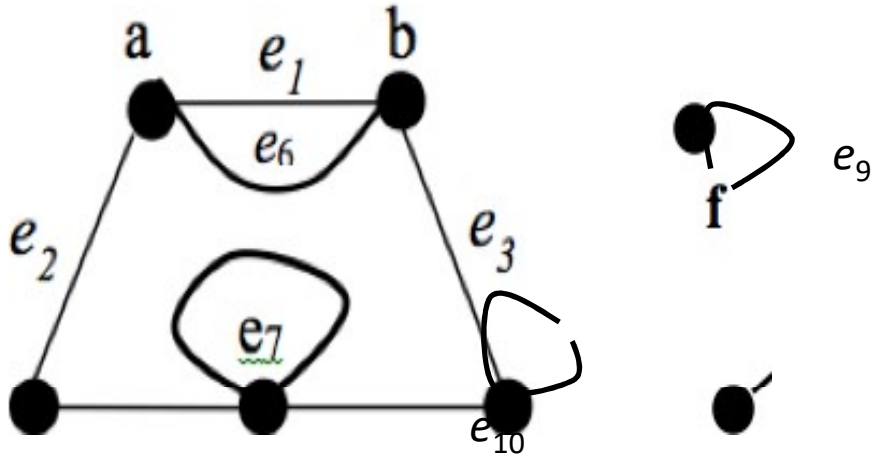


a) Write a vertex set and the edge set

$$V=\{a, b, c, d, e, f, h, i\}$$

$$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8, e_9, e_{10}\}$$

Example 2(cont)

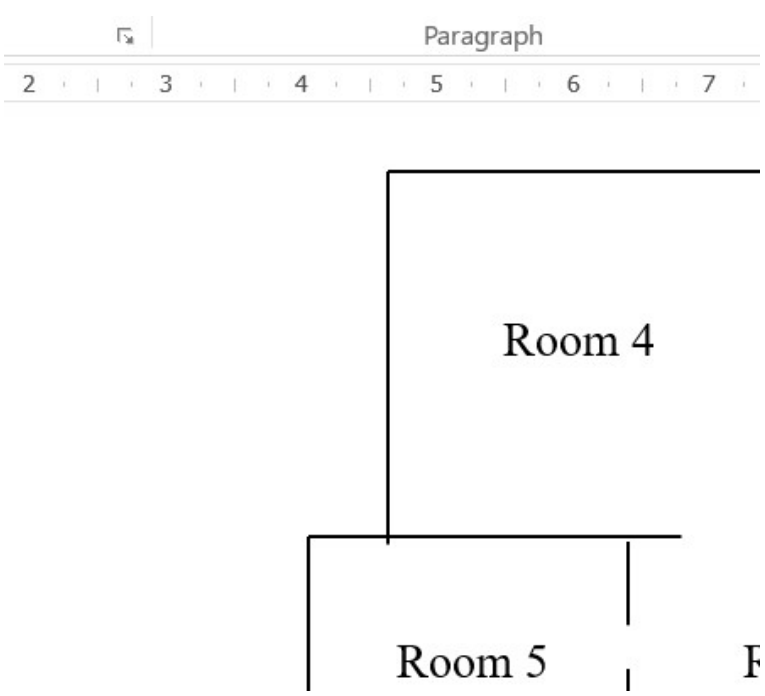


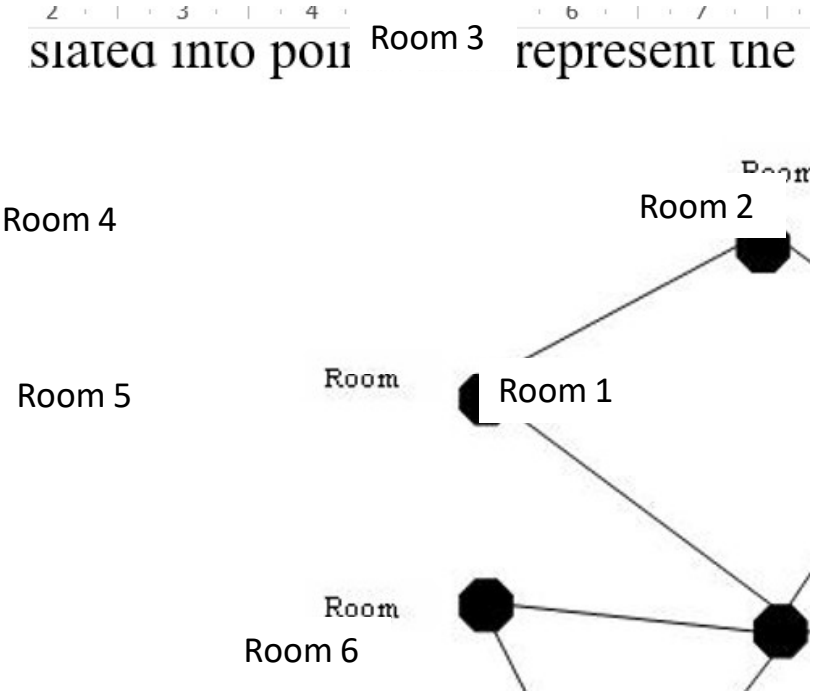
b) Give a table showing the edge-endpoint function.

Edge	Endpoints
e_1	{a, b}
e_2	{a, c}
e_3	{b, e}
e_4	{c, d}
e_5	{d, e}
e_6	{a, b}
e_7	{d}
e_8	{h, i}
e_9	{h}
e_{10}	{i}

$$f(e_1)=\{\mathbf{a}, \mathbf{b}\}, f(e_2)=\{\mathbf{a}, \mathbf{c}\}, f(e_3)=\{\mathbf{b}, \mathbf{e}\},$$
$$f(e_4)=\{\mathbf{c}, \mathbf{d}\}, f(e_5)=\{\mathbf{d}, \mathbf{e}\}, f(e_6)=\{\mathbf{a}, \mathbf{b}\},$$
$$f(e_7)=\{\mathbf{d}\}. f(e_8)=\{\mathbf{h,i}\}, f(e_9)=\{\mathbf{h}\},$$
$$f(e_0)=\{\mathbf{i}\}.$$

Application-Arrangement

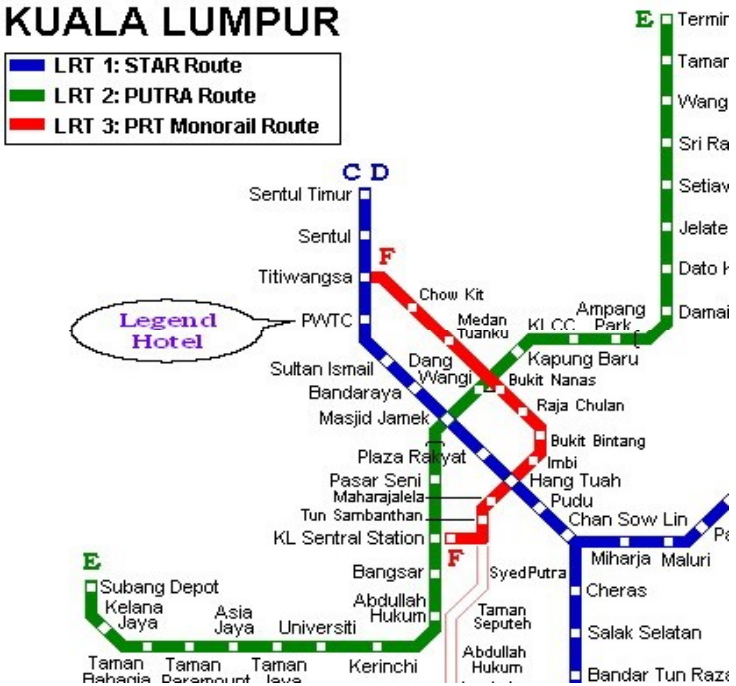




Vertices – rooms

Edges – door between the rooms.

Application-Map of the railway

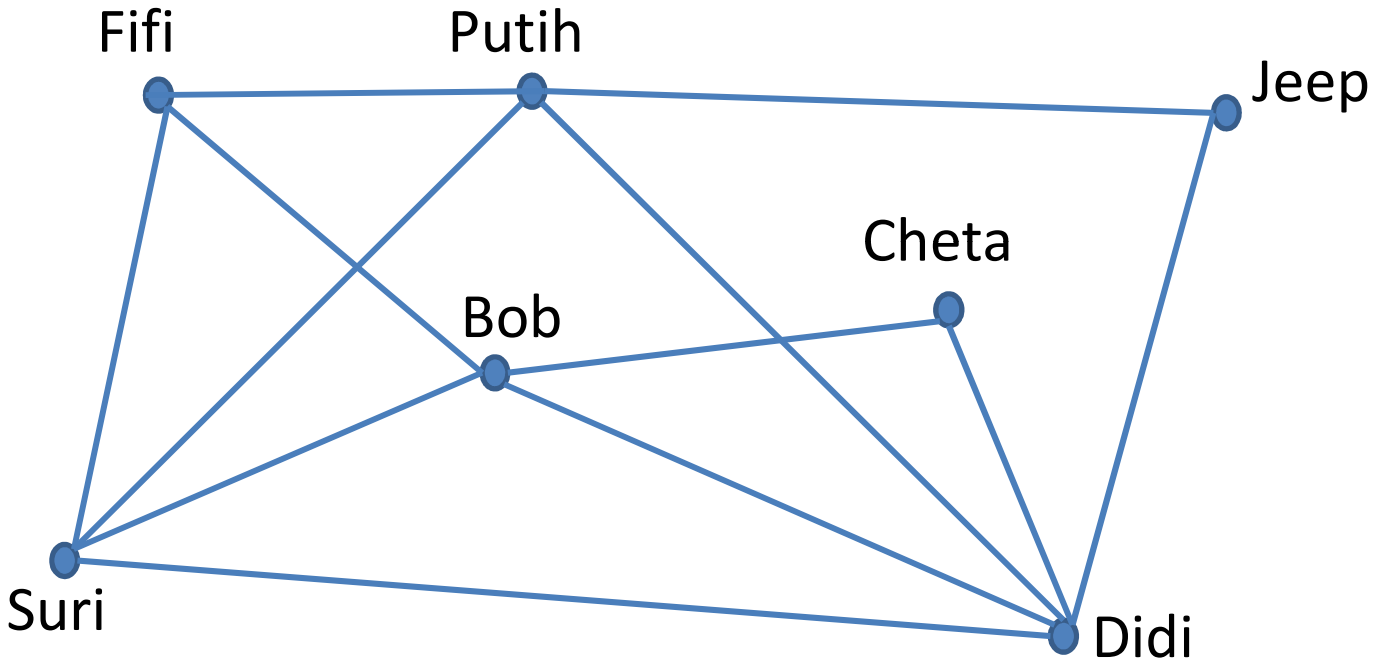


- Vertices – stations
- Edges - rail network

Past Year 2015/2016

a) A cat show is being judged from pictures of the cats. The judge pictures of the following pairs of cats next to each other for their fir Putih, Fifi and Suri, Fifi and Bob, Bob and Cheta, Bob and Didi, I and Didi, Didi and Suri, Didi and Putih, Suri and Putih, Putih and Je

Solution
Past Year 2015/2016

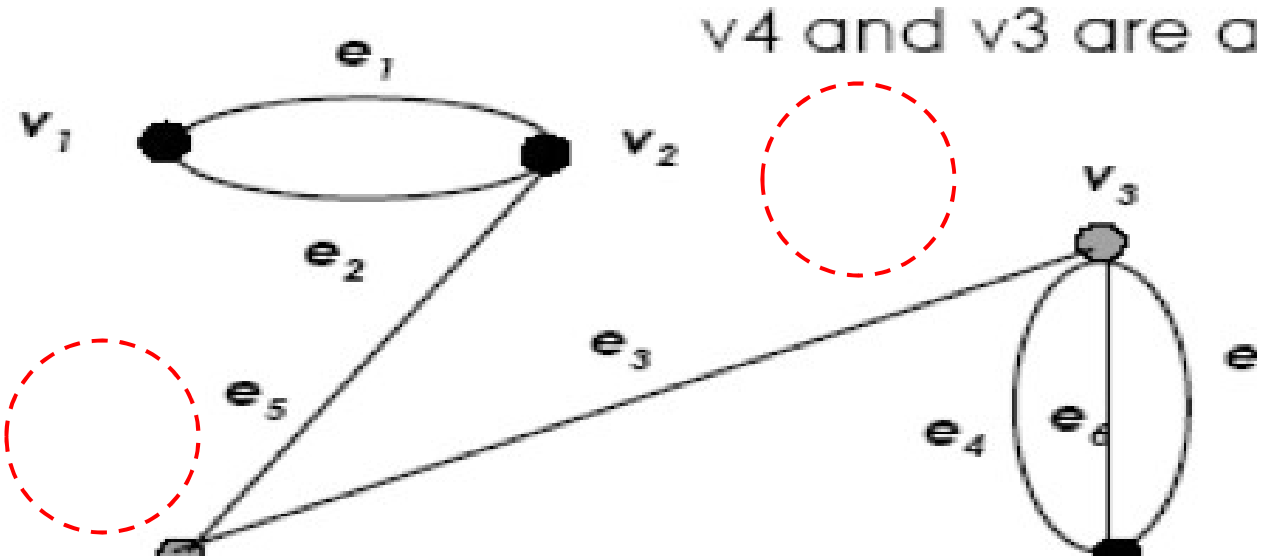


Characteristics of Graph

(a)Adjacent Vertices

Two vertices that are connected by an edge are called adjacent; and a vertex that is an endpoint of a loop is said to be adjacent to itself.

Example

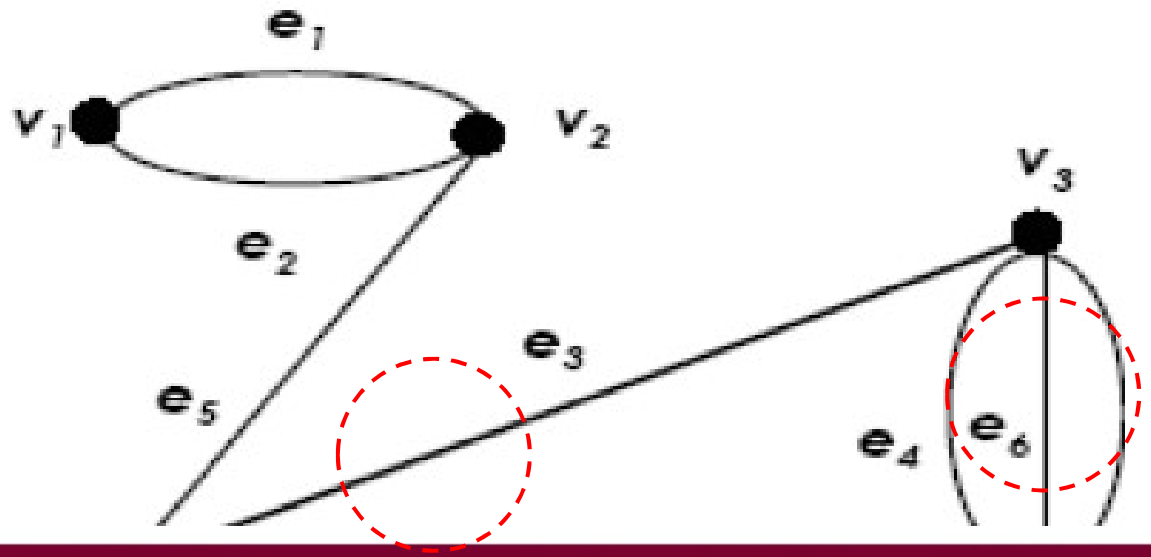


b) Isolated Vertex

Let G be a graph and v be a vertex in G. We say that v is an isolated vertex if it is not incident with any edge.

- v5 and v7 are isolated vertices.

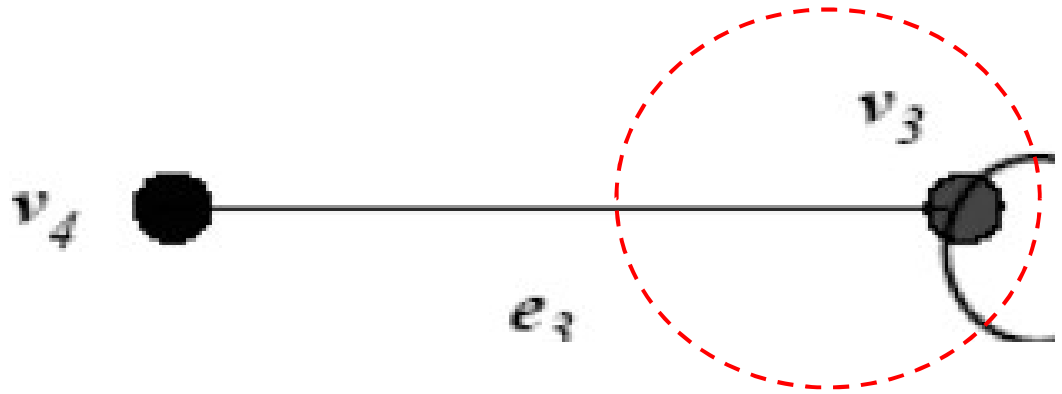
Example



c) Loop

An edge with just one endpoint is called a loop.

Example



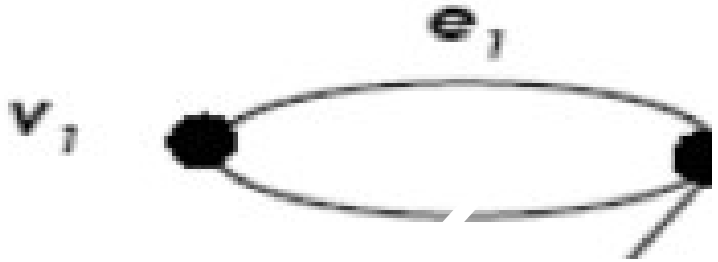
- e2 is a loc

d) Parallel Edges

Two or more distinct edges with the same set of endpoints are said to be parallel.

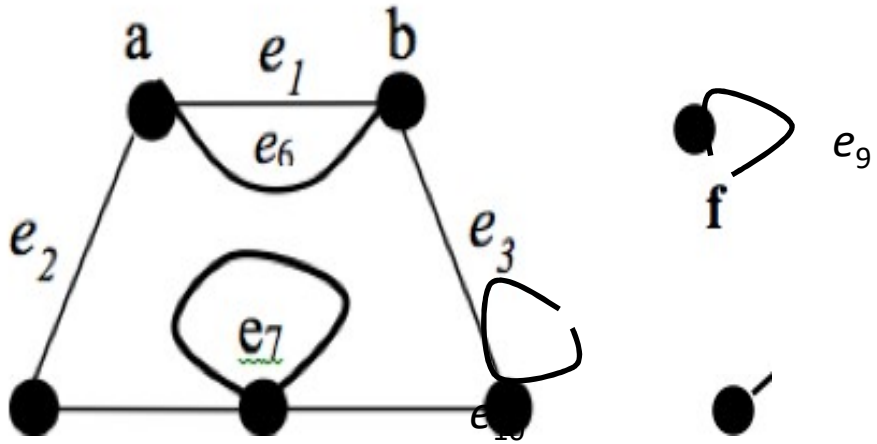
- e_1 and e_2 are parallel.

Example



Example 3

Given a graph as shown below,



Find all edges that are incident on **a**, all vertices that are adjacent to **a**, all edges that are adjacent to e_2, all loops, all parallel edges and all isolated vertices.

e_1, e_2 and e_6 are incident on **a**.

c and **b** are adjacent to **a**.

e_1, e_4 and e_6 are adjacent to e_2.

e_7 ,e_7 and e_{10} are loops.

e_1 and e_6 are parallel edges.

f is an isolated vertex.

The Concept of Degree

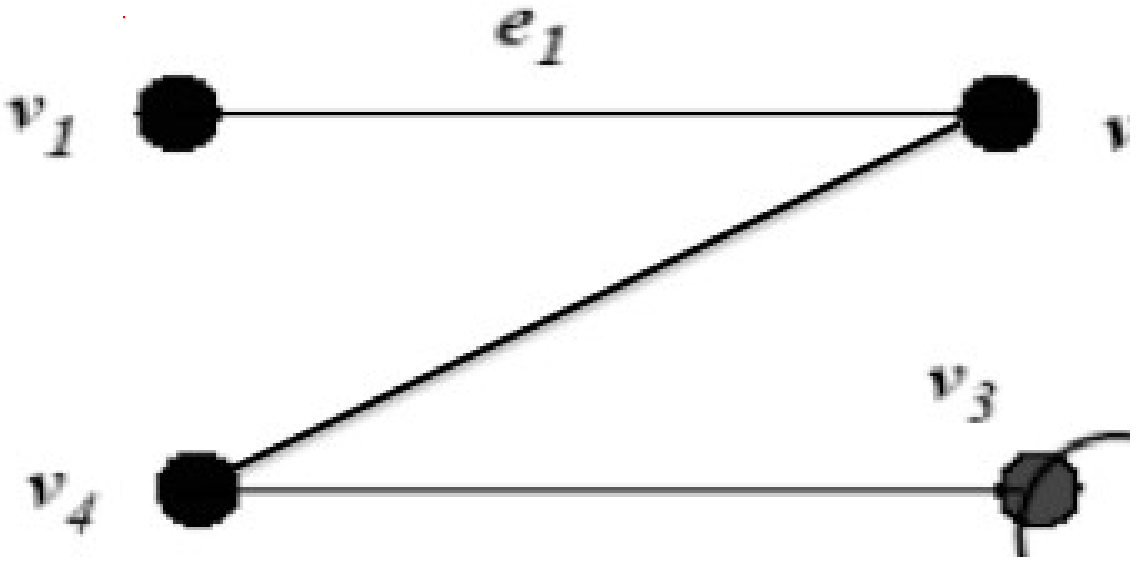
- Let G be a graph and v be a vertex in G.

- The degree of v, written deg(v) or d(v) is the number of edges incident with v.

- Each **loop** on a vertex v contributes 2 to the degree of v.

Example 4

State the degree of each vertex for the following graph.



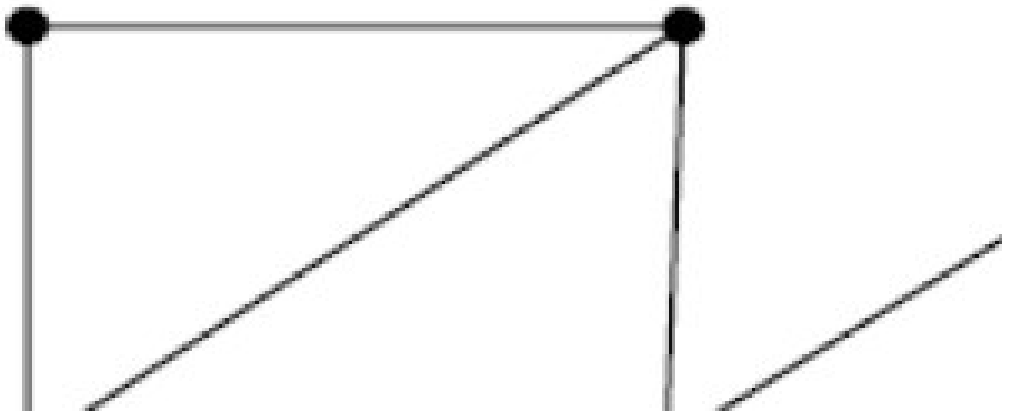
deg(v_1)= 1; deg(v_2)= 2; deg(v_3)= 3; deg(v_4)= 2

Types of Graphs

a) Simple Graph

A graph G is called a simple graph if G does not contain any parallel edges and any loops.

Example



b) Regular Graph

Let G be a graph and k be a nonnegative integer. G is called a k-regular graph if the degree of each vertex of G is k.

Example

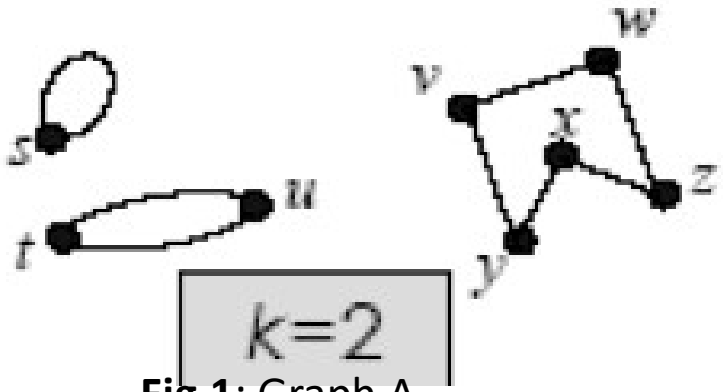


Fig.1: Graph A

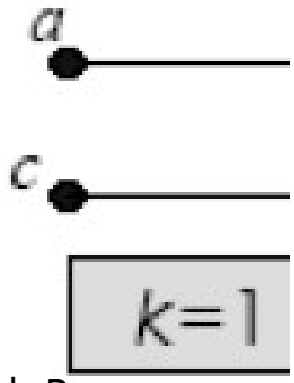


Fig.2: Graph B

c) Connected Graph

A graph G is connected if given any vertices v and w in G, there is a path from v to w.

Example

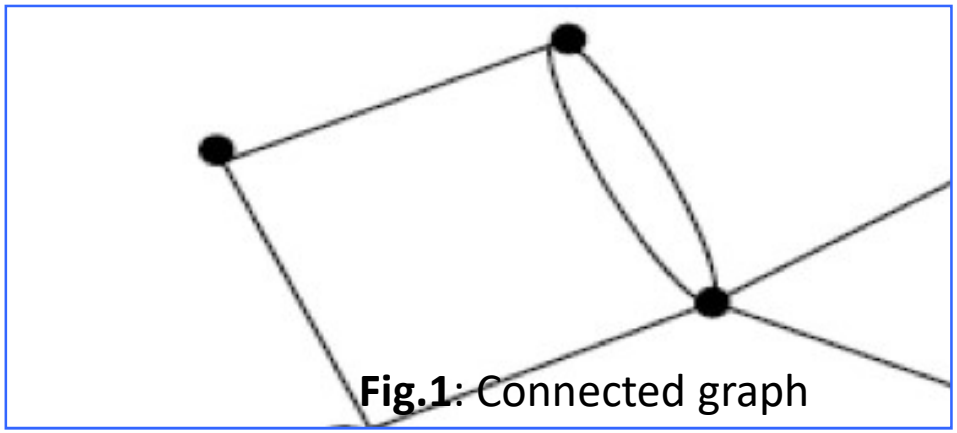


Fig.1: Connected graph

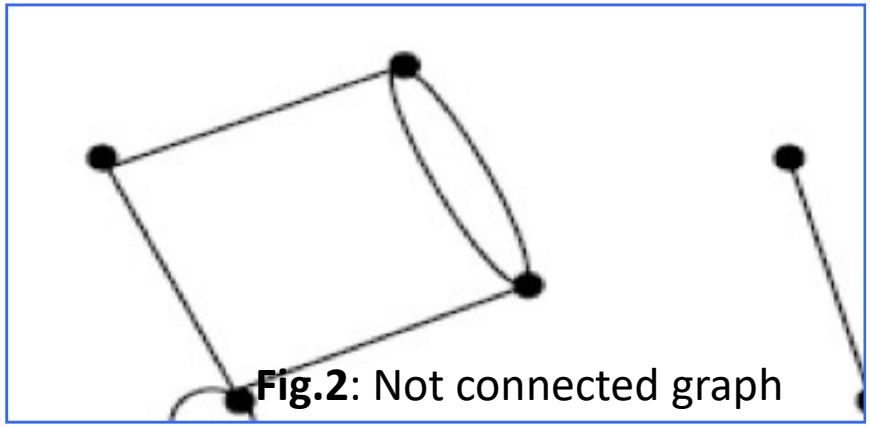


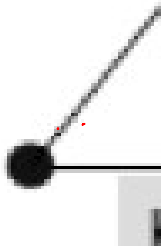
Fig.2: Not connected graph

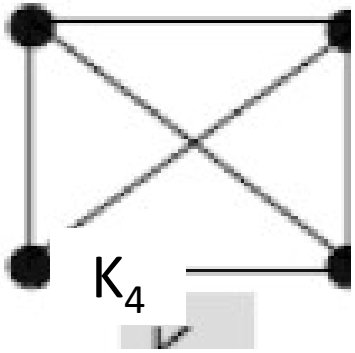
d) Complete Graph

A simple graph with n vertices in which there is an edge between every pair of distinct vertices is called a complete graph on n vertices. This is denoted by k_n .

Example



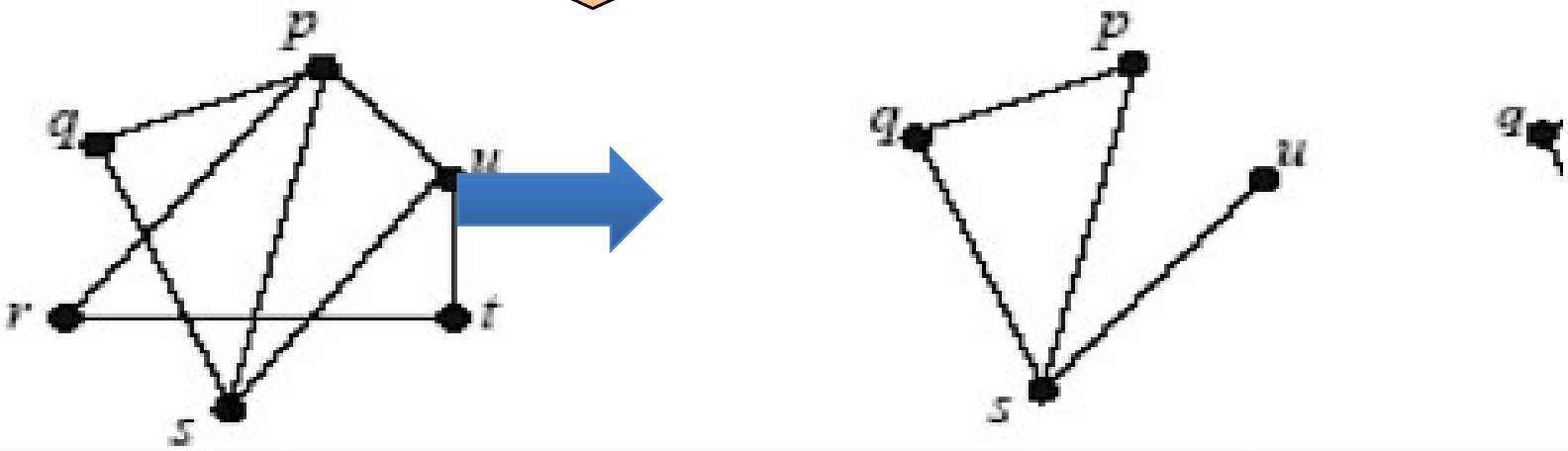




e) Subgraph

A graph H is said to be a subgraph of a graph G if, every vertex in H is also a vertex in G, every edge in H is also an edge in G, and every edge in H has the same endpoints as it has in G.

Example



Graph Representation

- To write programs that process and manipulate graphs, the graphs must be stored, that is, represented in computer memory.
- A graph can be represented (in computer memory) in several ways.
- 2-dimensional array: adjacency matrix and incidence matrix.

Adjacency Matrix

- Let G be a graph with n vertices.

- The adjacency matrix, A_G is an $n \times n$ matrix $[a_{ij}]$ such that,

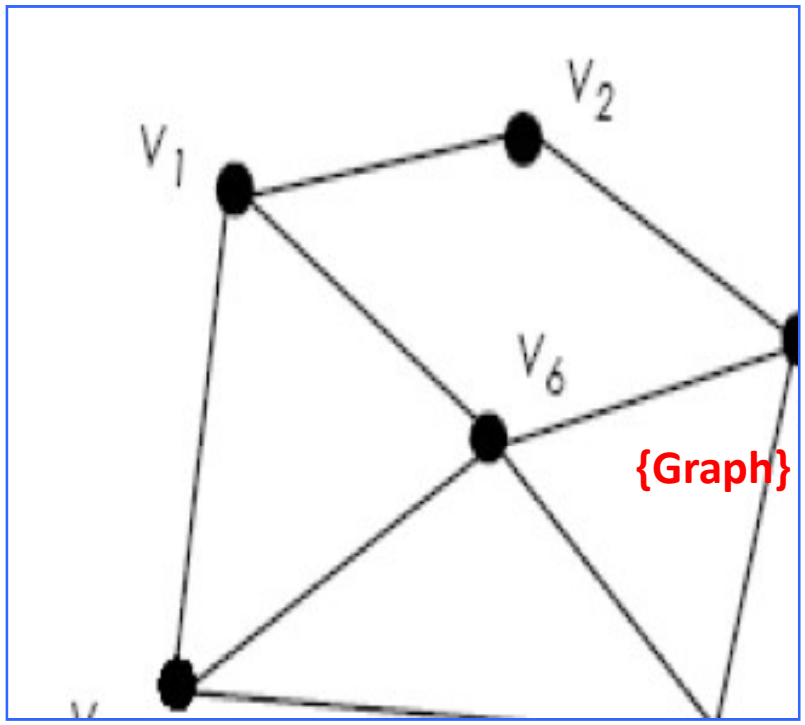
  a_{ij}= the number of edges from v_i to v_j, {undirected G}

  or,

  a_{ij}= the number of arrows from v_i to v_j, {directed G}

  for all i, j = 1,2,….., n.

Example 5



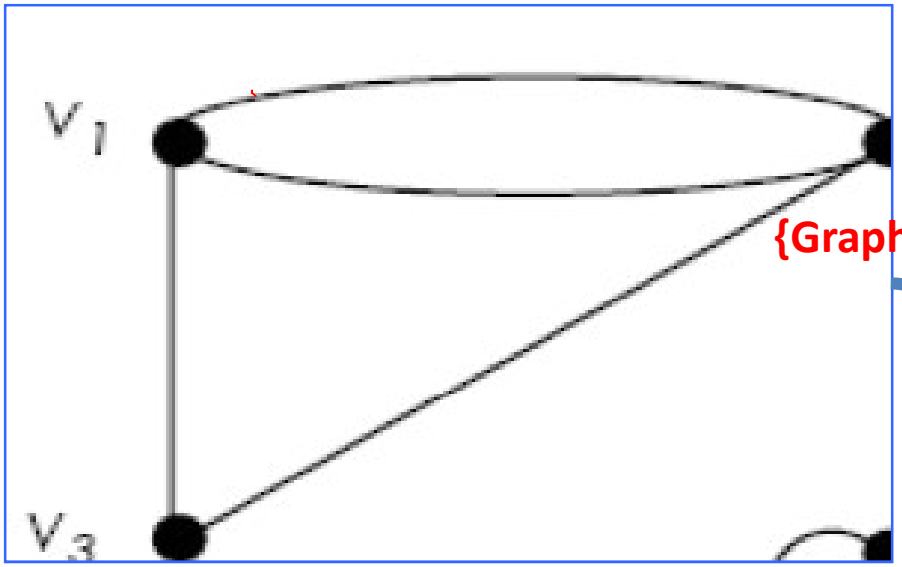
{Graph}

[Matrix]

$A_G =$

	v_1	v_2	v_3	v_4	v_5	v_6
v_1	0	1	0	0	1	1
v_2	1	0	1	0	0	0
v_3	0	1	0	1	0	1
v_4	0	0	1	0	1	1
v_5	1	0	0	1	0	1
v_6	1	0	1	1	1	0

Example 6



{Graph}

[Matrix]

$$A_G = \begin{array}{c} \begin{matrix} V_1 & V_2 & V_3 & V_4 \end{matrix} \\ \begin{bmatrix} 0 & 2 & 1 & 0 \\ 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

Example 7

Draw the graph based on the following matrix:

$$A_G = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

- Adjacency matrix is a **symmetric matrix** if it is representing an undirected graph, where

$$a_{ij} = a_{ji}$$

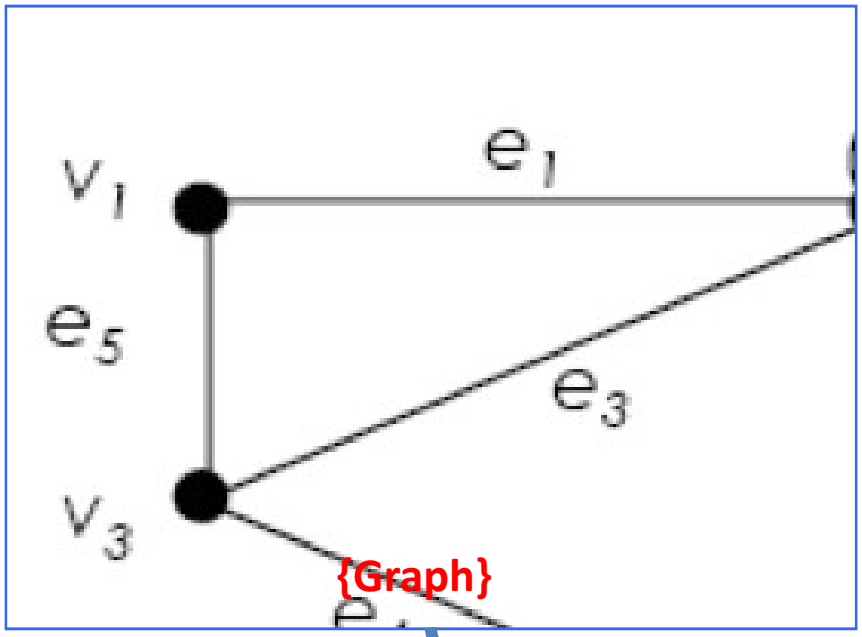
- If the graph is directed graph, the presented matrix is not symmetrical.

Incidence Matrix

- Let G be a graph with n vertices and m edges.
- The incidence matrix, I_G is an nxm matrix $[a_{ij}]$ such that,

$$a_{ij} = \begin{cases} 0 & \text{if } v_i \text{ is not an end vertex of } e_j, \\ 1 & \text{if } v_i \text{ is an end vertex of } e_j, \text{ but } e_j \text{ is not a loop} \\ 2 & \text{if } e_j \text{ is a loop at } v_i \end{cases}$$

Example 8



{Graph}

[Matrix]

$$\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \end{array} \begin{array}{c} \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & \end{array} \\ \left[\begin{array}{ccccc} 1 & 0 & 0 & 0 & 1 \\ 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{array}\right. \end{array}$$

$\deg(v_1) = 2;$
$\deg(v_2) = 4;$
$\deg(v_3) = 3;$
$\deg(v_4) = 1$

Notice that the sum of the i-th row is the degree of v_i.

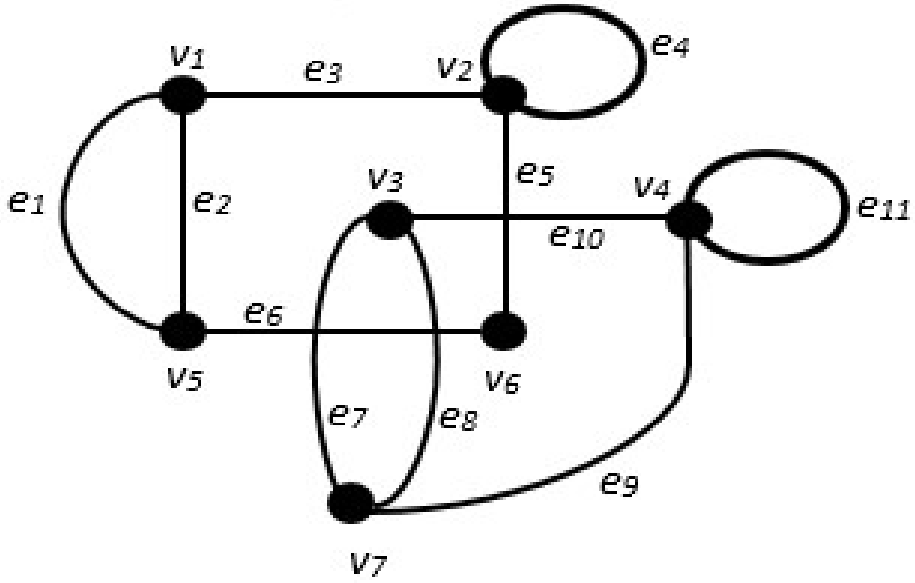
QUESTION 1

a) Model the following situations as graphs.

Aida and Bertrand are friends. Aida is also friends with Chong and David. Bertrand, Chong and Emilin are all friends of each other.

QUESTION 1b

b) Given the graph *H*



i)Construct the incidence matrix for graph *H*

ii)List the degree of each of the vertices

iii)Explain whether the graph is connected or disconnected?